Symmetrix-Polar architecture evolution and measured speed (2026-10-22)

(a) Baseline: paper / upstream

linear MACE, FP32, no global SCF, no exact Ewald

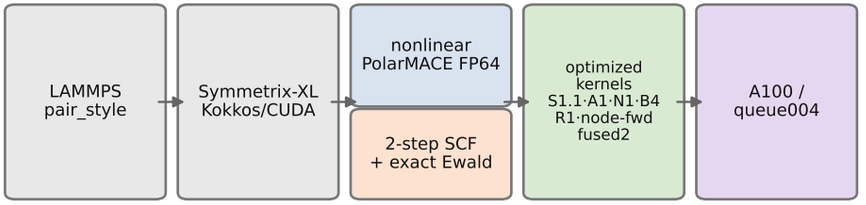
Measured on the paper's own workload:

4.50 µs/atom/step at 5,184 atoms

This is the matched streaming baseline; it does not run the polar model.

(b) Current: PolarMACE engine

same trained model; different execution path



Measured end-to-end (clean q4 A/B):

133.04 µs/atom/step at 1,296 atoms

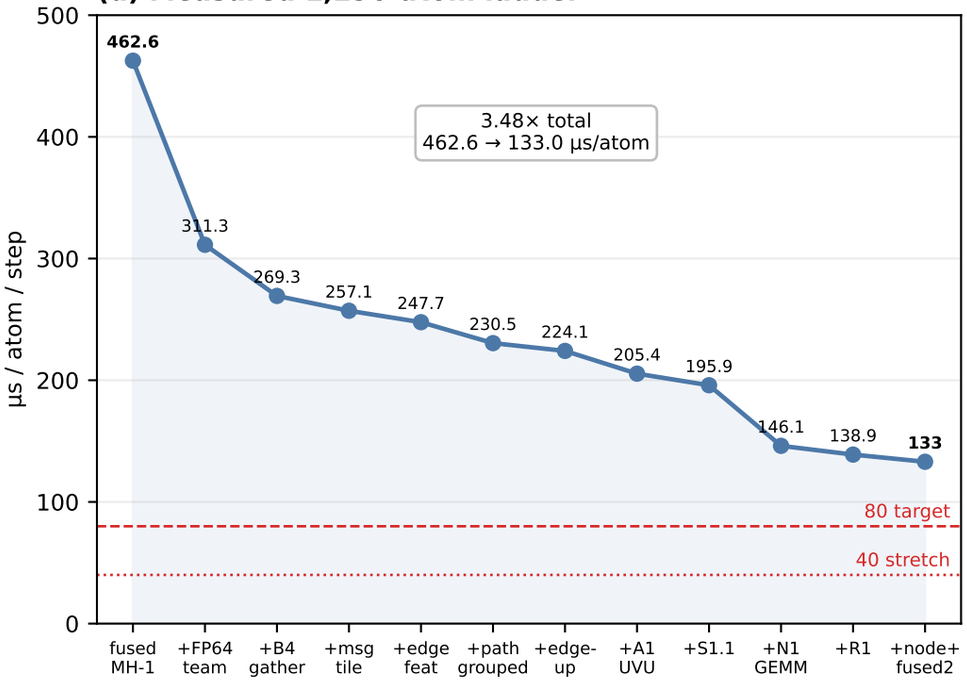
138.91 baseline → 133.04 (−4.2%); 140.91 at 5,184

(c) Implementation changes (model unchanged)

Change	Measured effect
S1.1 node-contiguous streamed-edge tiles + fused reverse	**253.84 ms @1,296; 8.55 GiB**
A1 UVU 32×16 tile, chunked weight staging	**290.20 → 266.21 ms (−8.26%)**
N1 grouped FP64 path-GEMM (6 tables, 20 planes)	**255.35 → 189.30 ms (−25.87%)**
Team MH-1 + B4 + message/edge/path tiles	**462.6 → 224.1 µs/atom**
V3 prepared path-plan cache	**~24 D2H syncs/eval removed**
R1 reverse MH-1 register accumulation	**−3.05% same-binary**
Node-owned forward dataflow	**−2.92 µs/atom (q4)**
Fused2 reverse dataflow	**−2.83 µs/atom (q4)**
q4 clean A/B, both default-on	**138.91 → 133.04 (−4.2%)**

Weights, equations, FP64, exact Ewald, and the LAMMPS interface are unchanged.

(d) Measured 1,296-atom ladder



Sources: docs/results/OPTIMIZATION_LEDGER.md; docs/results/LATEST_SPEED_AND_PLAN_2026-10-22.md; q4 A/B t-20261022180640-dslc9. The polar model's weights and equations are unchanged; all gains are execution changes.